CubeSat Low-Power Control Unit (LPCU)

AESS Sustainability Hackathon 2026 · Challenge 1: Sustainable Electronics for Space Systems · Team: cubesat

96.6%	×29.3	3.3%
Power reduction	Mission lifetime gain	Active duty cycle

1. Problem

Forest fires are accelerating globally. Space-based thermal monitoring offers wide-area, continuous coverage that ground sensors cannot match — but only if the satellite stays alive long enough to be useful. The limiting factor is power.

Most CubeSat housekeeping subsystems run in fixed always-on mode, drawing full current even when the mission task requires active processing for less than 4% of the operating period. In a 1U CubeSat with a 1–5 W total power budget, a housekeeping unit consuming 277 mW continuously starves the payload and shortens mission life — directly reducing the fire-detection capability the satellite exists to provide.

2. Solution

We designed a Low-Power Control Unit (LPCU) for a 1U CubeSat in Low Earth Orbit (500 km, 90-minute period). The LPCU performs attitude monitoring and health telemetry — essential housekeeping for a forest-fire detection mission — using an ESP32 microcontroller with hardware deep sleep and a strict 3.3% duty cycle.

Power-saving strategy:

Technique	How it works	Saving
ESP32 deep sleep	ULP timer wakes main cores every 60 s	255.2 mW
MPU-6050 cycle mode	IMU standby between measurements (5 µA)	10.8 mW
TMP117 one-shot mode	Sensor active only during 15.5 ms conversion	1.6 mW
Low-Iq regulator	TPS63021: 60 µA quiescent, buck-boost	baseline
Total saving	Duty cycle D = 2 s active / 60 s cycle	267.6 mW

3. Power budget

Component	Baseline (mW)	Optimized (mW)	Reduction
ESP32 MCU	264.0	8.8	96.7%
MPU-6050 IMU	11.2	0.4	96.4%
TMP117 temp.	1.7	0.1	94.1%
TPS63021 vreg.	0.2	0.2	—
TOTAL	**277.1**	**9.5**	**96.6%**

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P_avg = P_active × D + P_sleep × (1 − D) → ESP32: 264.0 × 0.033 + 0.033 × 0.967 = 8.79 mW
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4. Implementation

Deliverable	Description
Python simulation	power_model.py — reproduces all power numbers, outputs power_results.json
ESP32 firmware	lpcu_firmware.ino — real deep sleep cycle, I2C sensor reads, UART telemetry
1U prototype	Physical hardware: ESP32 + MPU-6050 + TMP117, flashed and operating
Validation	Serial monitor confirms 2 s active / 58 s sleep cycle in real hardware

5. Sustainability impact

Metric	Baseline	Optimized	Impact
Avg. system power	277.1 mW	9.5 mW	267 mW freed for payload
Energy per orbit	0.416 Wh	0.014 Wh	96.6% less per orbit
Battery runtime (1 Wh)	0.06 days	1.76 days	Eclipse anomaly survival
Mission data per launch	1×	29.3×	Better lifecycle value

For a forest-fire detection CubeSat, the 267 mW freed by the LPCU can power a thermal infrared sensor node — the actual fire-detection payload. Every milliwatt saved in housekeeping is a milliwatt available for the mission that justifies the launch.